Experiment: Characterization of a Power Combiner

Objective

To characterize the combining response of a two-input RF power combiner by applying equal input powers and measuring the output power at different frequencies.

Components Required

RF signal generators (2 Nos.), RF power combiner, power meter/spectrum analyzer, RF cables and connectors, and suitable 50 Ω termination/load.

Theory

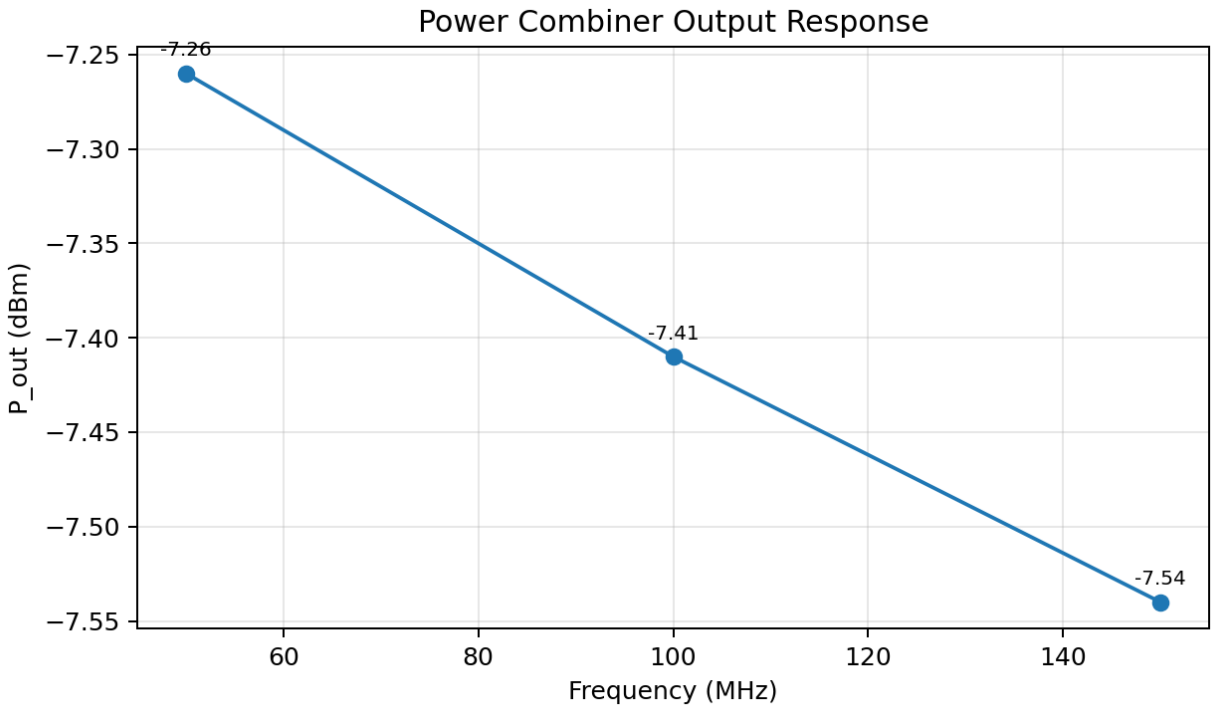
A power combiner is an RF/microwave device used to combine two or more input signals into a single output. For two equal input powers, the ideal combined power is obtained by adding the powers in linear units. Here, P_{in1} = –3 dBm and P_{in2} = –3 dBm. The ideal combined power is 0 dBm. The practical P_{out} is lower because of insertion loss, mismatch and other losses.

Experimental Setup

Two RF signal sources were connected to the two input ports of the power combiner. Both input powers were maintained at –3 dBm. The output port was connected to the measuring instrument, and P_{out} was measured at 50 MHz, 100 MHz and 150 MHz.

Observation & Graphs

Input Frequency (MHz)	P_in1 (dBm)	P_in2 (dBm)	P_out (dBm)	IL (dB)
50	-3.00	-3.00	-7.26	7.26
100	-3.00	-3.00	-7.41	7.41
150	-3.00	-3.00	-7.54	7.54



Calculation & Results

Input Powers:

P_{in1} = –3 dBm

P_{in2} = –3 dBm

Ideal combined output:

P_{ideal} = 0 dBm

Insertion Loss:

$IL = P_{ideal} - P_{out}$

At 50 MHz: IL = 0 – (–7.26) = **7.26 dB**

At 100 MHz: IL = 0 – (–7.41) = **7.41 dB**

At 150 MHz: IL = 0 – (–7.54) = **7.54 dB**

Conclusion (Key Point / Takeaway)

The two-input RF power combiner was successfully characterized using equal input powers of –3 dBm. The measured P_{out} values were –7.26 dBm, –7.41 dBm and –7.54 dBm at 50 MHz, 100 MHz and 150 MHz, respectively. The practical output is lower than the ideal combined power because of insertion loss and other RF losses.